

ICMP Lab

The image shows a Wireshark packet capture of ICMP Echo (ping) traffic. The packet list pane shows multiple requests and replies. The packet details pane for packet 2371 shows the ICMP Echo (ping) request details. The packet bytes pane shows the raw data of the ICMP packet. A Windows PowerShell window is overlaid on the right, showing the output of a ping command to www.ust.hk.

0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0x268d (9869)
000. = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.20.11.119
Destination Address: 143.89.209.9
[Stream index: 315]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0

Pinging www.ust.hk [143.89.209.9] with 32 bytes of data:
Reply from 143.89.209.9: bytes=32 time=221ms TTL=35
Reply from 143.89.209.9: bytes=32 time=227ms TTL=35
Reply from 143.89.209.9: bytes=32 time=398ms TTL=35
Reply from 143.89.209.9: bytes=32 time=470ms TTL=35
Reply from 143.89.209.9: bytes=32 time=377ms TTL=35
Reply from 143.89.209.9: bytes=32 time=414ms TTL=35
Request timed out.
Reply from 143.89.209.9: bytes=32 time=219ms TTL=35
Reply from 143.89.209.9: bytes=32 time=241ms TTL=35
Reply from 143.89.209.9: bytes=32 time=257ms TTL=35
Ping statistics for 143.89.209.9:
Packets: Sent = 10, Received = 9, Lost = 1 (10% loss),
Approximate round trip times in milli-seconds:
Minimum = 219ms, Maximum = 470ms, Average = 313ms
PS C:\Users\bhavad>

1. Your host (source) IP: 10.20.11.119
Destination host IP: 143.89.209.9 (www.ust.hk)
2. ICMP does not have source/destination port numbers because:
 - a. ICMP is a Layer 3 (Network Layer) protocol, sitting alongside IP — not above it like TCP/UDP
 - b. Port numbers are a Layer 4 (Transport Layer) concept used by TCP and UDP to direct traffic to specific applications
 - c. ICMP's purpose is to send network-level error and control messages, not application data, so it has no need for ports
 - d. Notice in my capture the field says Protocol: ICMP (1), it goes directly into the IP payload, bypassing the transport layer entirely
3. ICMP Type and Code:
 - a. Type: 8 → Echo (ping) request
 - b. Code: 0 → No subtype, just a standard echo request

Other fields in this ICMP packet:

Field	Value	Size
Checksum	0x4d05	2 bytes
Identifier	0x0001	2 bytes
Sequence Number	86 (0x0056)	2 bytes
Data Payload	6162636465...	32 bytes

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4. ICMP Type and Code:
 - a. Type: 0 → Echo (ping) reply
 - b. Code: 0 → Standard echo reply

Other fields in this ICMP packet:

Field	Value	Size
Checksum	0x5505	2 bytes
Identifier	0x0001	2 bytes
Sequence Number	86 (0x0056)	2 bytes
Data Payload	6162636465... 32 bytes	

The image shows a Wireshark packet capture of ICMP Echo (ping) requests. The packet list shows multiple requests from 10.20.11.119 to 128.93.162.83. The packet details pane shows the ICMP Echo (ping) request details, including the sequence number 86 and the data payload. A Windows PowerShell window is open in the foreground, showing the command `tracert www.inria.fr` and its output, which displays the route from 10.20.11.119 to 128.93.162.83.

```
PS C:\Users\l\hnavd> tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  10.20.0.1
  1  3 ms  3 ms  9 ms  192.168.46.1
  2  5 ms  26 ms  13 ms  192.168.45.9
  3  22 ms  7 ms  4 ms  205.189.187.3
  4  9 ms  6 ms  6 ms  66-207-208-113.beanfield.net [66.207.208.113]
  5  *  *  *  Request timed out.
  6  *  *  *  Request timed out.
  7  112 ms  25 ms  128 ms  et33-1.lsr01.3389SteelesAve01.yyz.beanfield.com [207.66.123.175]
  8  230 ms  164 ms  190 ms  be320.dr01.3389SteelesAve01.yyz.beanfield.com [199.167.154.210]
  9  *  *  *  Request timed out.
 10  *  *  *  Request timed out.
 11  *  *  *  Request timed out.
 12  10 ms  11 ms  6 ms  beanfield-ic-357184.ip.twelve99-cust.net [80.239.133.81]
```

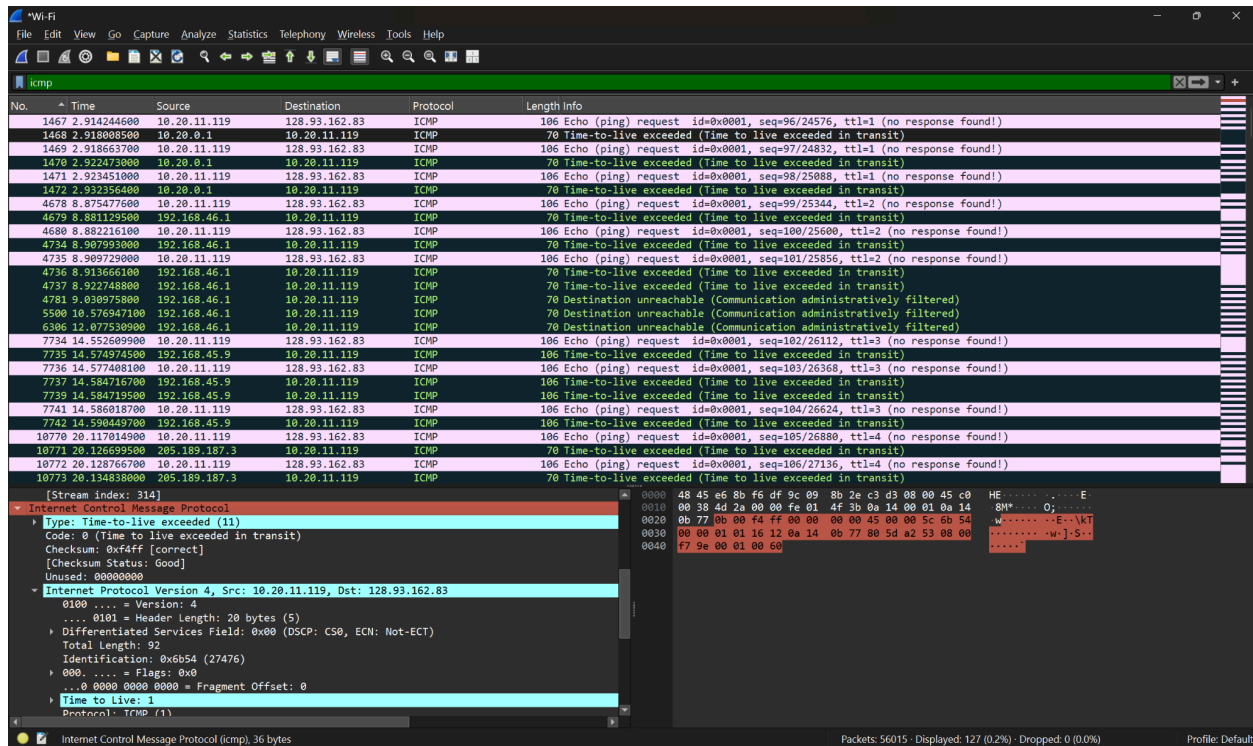
5. Your host (source) IP: 10.20.11.119 (same as before)
Target destination IP: 128.93.162.83 (www.inria.fr — France)
6. Windows traceroute uses ICMP → Protocol number = 1
If it sent UDP packets instead (like Unix/Linux does), the protocol number would be 17 (0x11), So no, it would NOT still be 01
7. Comparison between Ping and Traceroute packets:

Field	Ping Request	Traceroute Request
Type	8	8 same
Code	0	0 same
Identifier	0x0001	0x0001 same
Sequence Number	86	96
TTL (IP Header)	128	1 different
Size	60 bytes	106 bytes

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Key difference: TTL value in the IP header

- Ping: Sends packets with TTL = 128 (normal value, intended to reach the destination directly).
- Traceroute: Starts with TTL = 1 and increases it each round. This forces routers along the path to send back TTL-exceeded error messages, allowing traceroute to discover the location of each router between the source and destination.



- The TTL-exceeded error packet contains two layers inside it:
Outer ICMP Layer (Error message from router 10.20.0.1)

Field	Value
Type	11 (Time-to-live exceeded)
Code	0 (TTL exceeded in transit)
Checksum	0xf4ff
Unused	00000000 (4 bytes, padding)

- Embedded Original IP Header (the packet that caused the error)

Field	Value
Source	10.20.11.119 (your host)
Destination	128.93.162.83 (inria.fr)
TTL	1 ← this is why it was dropped
Protocol	ICMP (1)

3. Original ICMP Header (the probe that triggered the error)

Field	Value
Type	8 (Echo request)
Sequence Number	96
Identifier	0x0001

The router includes the original IP header + original ICMP header inside the error message so the sender can identify exactly which probe caused the TTL error. Without this information, your host would not be able to match the TTL-exceeded error back to the correct traceroute request.

9. Comparison Between Final Traceroute Packets and TTL-Exceeded Error Packets

Last 3 packets:

Packet	From	Type	Sequence Number
162	128.93.162.83	Echo Reply (Type 0)	163
163	10.20.11.119	Echo Request (Type 8)	164
164	128.93.162.83	Echo Reply (Type 0)	164

How are they different from TTL-exceeded error packets?

Feature	TTL-Exceeded Packets	Last 3 Packets
Type	11 (Error)	0 (Echo Reply)
Sent by	Intermediate routers	Destination (inria.fr) itself
Contains embedded header?	Yes	No
TTL of reply	Low (nearby routers)	39 (came from far away)
Meaning	"Your packet died here"	"I received your ping!"

Why they're different: Traceroute eventually sent probes with a high enough TTL value to reach inria.fr in France. Instead of a router dropping the packet and sending back a Type 11 TTL-exceeded error, the destination host received the ping and responded with a normal Echo Reply (Type 0).

This shows that the traceroute has reached the final destination and the route is complete.

```

Windows PowerShell
PS C:\Users\bhavd> tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

  1    3 ms    3 ms    9 ms  10.20.0.1
  2    5 ms   26 ms   13 ms  192.168.46.1
  3   22 ms    7 ms    4 ms  192.168.45.9
  4    9 ms    6 ms    6 ms  205.189.187.3
  5    *   169 ms  208 ms  66-207-208-113.beanfield.net [66.207.208.113]
  6    *      *      *      Request timed out.
  7  112 ms   25 ms  128 ms  et33-1.lsr01.3389SteelesAvE01.yyz.beanfield.com [207.66.123.175]
  8  230 ms  164 ms  190 ms  be320.dr01.3389steelesave01.yyz.beanfield.com [199.167.154.210]
  9    *      *      *      Request timed out.
 10   *      *      *      Request timed out.
 11   *      *      *      Request timed out.
 12   10 ms   11 ms    6 ms  beanfield-ic-357184.ip.twelve99-cust.net [80.239.133.81]
 13    6 ms   13 ms    8 ms  toro-b5-link.ip.twelve99.net [80.239.133.80]
 14   12 ms   11 ms   13 ms  toro-b4-link.ip.twelve99.net [62.115.141.34]
 15   *      *      *      Request timed out.
 16   *      44 ms   36 ms  nyk-b17-link.ip.twelve99.net [62.115.137.15]
 17   26 ms   30 ms   57 ms  telecomitalia-ic-364638.ip.twelve99-cust.net [80.239.135.165]
 18  100 ms  116 ms  114 ms  195.22.210.95
 19  113 ms  120 ms  121 ms  213.144.183.207
 20  184 ms  267 ms  178 ms  hu0-3-0-41-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.131]
 21  602 ms   *   119 ms  inria-rocquencourt-vl1631-te1-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 22  112 ms   *   143 ms  unit240-reth1-vfw-ext-dc1.inria.fr [192.93.122.19]
 23  248 ms  286 ms  182 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.
PS C:\Users\bhavd>

```

10. Looking at the traceroute results, the main RTT jumps are:

Hop	Router	Average RTT	RTT Jump
7	beanfield ... yyz ...	~88 ms	—
8	be320.dr01 ... yyz ...	~195 ms	+107 ms
12	beanfield-ic ... twelve99 ...	~9 ms	—
16	nyk-b17 ...	~40 ms	—
17	telecomitalia ...	~38 ms	—
18	195.22.210.95	~110 ms	+72 ms
20	ren-nr-orsay...renater.fr	~210 ms	+90 ms
21	inria-rocquencourt...renater.fr	~602 ms	+392 ms

Most Significant Delay

The largest delay increase occurs between hops 20 and 21:

Hop 20: ren-nr-orsay-rtr-091.noc.renater.fr → Orsay, France

Hop 21: inria-rocquencourt...renater.fr → Rocquencourt, France

Hop 21 shows a large RTT spike of approximately 602 ms. This is likely caused by congestion, queueing, or packet loss on the RENATER research network rather than a long physical distance, since both hops are located in France.

The * on the second probe suggests possible instability or packet loss at that hop.

More Meaningful Geographic Delay

The more significant geographic transition occurs between:

Hop 16: nyk-b17-link.ip.twelve99.net → New York, USA

"nyk" indicates New York

Hop 17: telecomitalia-ic... → Italy/Europe

Traffic is handed off to Telecom Italia in Europe

This is where the traffic crosses the Atlantic Ocean from North America to Europe, which explains the sustained RTT increase of around 100 ms or more starting from hop 17 onward.